MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 1K3A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode, vacuum, high-voltage rectifier

Outline --- See figure 1

Envelope --- T9

Base --- Octal (see figure 1)

Cap --- C1-34 or C1-49

Cathode --- Coated filament (helical electrostatic shield - mandatory)

Base connections: See note 1

Pin No.	---	1	2	3	4	5	6	7	8	Cap
Element	---	int con	f	int con	nc	int con	nc	f	int con	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	epx	ib	Ic	tp	Du	Ⓒ X-radiation
Unit:	Vac	kv	ma	mAdc	μs (see note 2)	Note 2	mr/hr
Maximum	1.45	26 (see note 3)	50	0.5	10	0.15	0.5 (see note 4)
Minimum:	1.05	---	---	---	---	---	---
TEST CONDITIONS:	1.25	---	---	---	---	---	---

GENERAL:

Qualification - Not required

Ⓒ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1353	Operation of rectifiers	5,6	Rt = 100 Meg; Ct = 500 pF; prr = 15,750 (approx); tp = 10 µs (approx)	0.65	II	Eo	18.8	---	kVdc
1231	Emission (1)	-	Eb = 450 Vac	0.65	II	Is	4.0	---	mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Filament current	-		2.5	S3	If	180	220	mAdc
1231	Emission (2)	-	Eb = 450 Vac; Ef = 0.9 Vdc	2.5	S3	Is	3.2	---	mAdc
(C) 2202	X-radiation	-	Ef = 0; Eb = -26 kVdc	4.0	Code B	X-rad	---	0.5	mr/hr
(C) 1111	Base pin solder depth	-		---	---	---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group A; Ef = 1.25 V; Eo = 22 kV; Ri = 40 Meg; Ct = 500 pF; epk = 26 kv (approx); prr = 15,750 (approx); tp = 10 µs (approx)	---	---	---	---	---	---
---	Life-test end points (500 hours):								
1231	Emission (2)	-		---	---	Is	2.1	---	mAdc
(C) 1353	Operation of rectifiers	-		---	---	Eo	16.0	---	kVdc
(C) 2202	X-radiation	-		---	---	X-rad	---	0.5	mr/hr

NOTES:

1. Connecting pins 1, 3, 4, 5, 6, and 8 to pin 7 externally is permissible to reduce corona discharge. Otherwise, the connecting pins shall not be used.
2. tp shall be measured at the base of a pulse approximately half-sinusoidal in shape with less-than-critical damping. It is synonomous with the retrace time of the associated deflection circuit. Du is the product of pulse repetition rate and retrace time.
3. The dc component of the peak inverse anode voltage shall not exceed 22,000 volts.
4. (C) Based on accumulated sample test data, taken initially and during life test, tubes do not exceed this maximum rating limit of 0.5 mr/hr at any time throughout their useful life when operated within the absolute maximum ratings, including filament voltage, specified herein.
5. The duration of this test shall not exceed 1 minute. A tube that shows a temporary or permanent interruption in Eo or is not within specified limits for Eo during test shall be cause for rejection.
6. In a flyback pulse circuit, pulse voltage shall be adjusted so that a bogey tube gives Eo = 22,000 V. A bogey tube has a tube drop of Etd = 225 Vdc with Ik = 7.0 mAdc.

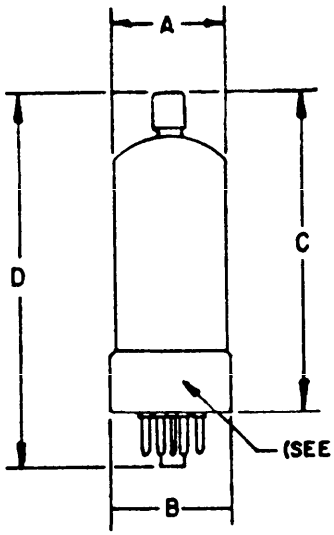
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

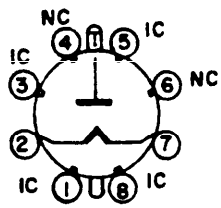
Agent:
DSA - ES

(Project 5960-N052)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	1.062 (26.97)	1.188 (30.18)
B		1.377 (34.98)
C	2.625 (66.68)	3.000 (76.20)
D		3.563 (90.50)



BASING DIAGRAM
(SEE NOTE b)

NOTES:

a. Base shall be one of the following:

B6-8 - Intermediate shell octal 6-pin or,
B8-6 - Intermediate shell octal 8-pin or,
B6-120 - Short medium-shell octal 6-pin or,
B8-118 - Short medium-shell octal 8-pin or,
B6-253 - Ultra-short small-wafer octal 6-pin or,
B8-251 - Ultra-short small-wafer octal 8-pin.

b. Pins 4 and 6 may be omitted.

FIGURE 1. Outline drawing of electron tube type 1K3A.

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